

PURUSHOTTAM KUMAR

+91-8757422328 | 772purushottam@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

EDUCATION

National Institute of Technology, Patna

B.Tech, Electronics and Communication Engineering | CGPA: 8.45/10

Jun. 2024 – May 2028

Patna, Bihar, India

PROJECTS

Multi-Threaded Proxy Server with LRU Cache

C, Socket Programming, POSIX Threads | [GitHub](#)

- Built a multi-threaded HTTP proxy server in C over raw TCP sockets, parsing and forwarding client requests to destination web servers.
- Engineered concurrent request handling with POSIX threads, semaphores, and mutex locks, eliminating race conditions across shared connections and cache state.
- Designed an LRU cache layer to store and reuse prior responses, avoiding redundant network calls on repeated requests.
- Built parallel implementations with and without caching to directly measure and demonstrate caching's impact on response latency.
- Applied core OS concepts — multithreading, synchronization, semaphores, locks, cache management — to a systems-level networking problem.

Full-Stack URL Shortener

React, Node.js, Express.js, REST API | [GitHub](#)

- Built a full-stack URL shortening platform with React, Node.js, and Express, letting users generate, manage, and track custom short links.
- Implemented shortcode validation, duplicate detection, and configurable link expiration on an in-memory data store.
- Exposed a dedicated API endpoint for on-demand QR-code generation, giving every shortened link a scannable QR counterpart.
- Built reusable React components — auth, link creation, analytics, QR display — with client-side validation for a consistent UX across the app.

ACHIEVEMENTS & CERTIFICATIONS

- Semi-Finalist, Flipkart GRiD 8.0 — advanced past initial screening rounds among 1.7 lakh+ applicants nationwide.
- Solved 1,000+ DSA problems across LeetCode, CodeChef, and Codeforces, reaching a LeetCode max rating of 1670 ([profile](#)), CodeChef 2-star, 1464 ([profile](#)), and Codeforces 970 ([profile](#)).
- Secured Elite standing (Top 5%) with a 96% score in NPTEL's "The Joy of Computing using Python," IIT Madras.
- Completed "VLSI SoC Design using Verilog HDL" certification from Maven Silicon, Centre of Excellence in Semicon.
- Achieved 98.6 percentile (All India Rank 22,255) in JEE Main 2024 and qualified JEE Advanced 2024.

TECHNICAL SKILLS

Languages: C++, Java, Python, C

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems

Frontend: React.js

Backend: Node.js, Express.js

Tools: Git, GitHub, IntelliJ IDEA, VS Code